

How modern MCUs achieve even lower power consumption

MCUs have dramatically dropped their RUN currents, allowing themselves to run more often, thereby allowing more features to be designed into the application software. In the past 18 months, we have seen MCUs with low-power libraries, gate-level optimisation and power gating on a rise. Performance-per-watt levels have evolved in many embedded applications to enable power-efficient designs, which help end users to conserve energy. Energy trends and power management are thus the key technology trends that designers are looking for nowadays.

—Sanjay Gupta, director, automotive MCU, India Design Centre, Freescale Semiconductor India Pvt Ltd

Key ingredients to a successful IoT recipe

- ▶ Low power performance (mainly for a battery-operated device)
- ▶ Medium to high D-MIPS performance (to operate under real-time constraints, some medium to high embedded complex software)
- ▶ Enough communication interface (SPI, I2C, USART) to connect to various sensors
- ▶ Display controllers (LCD/TFT/MIP)
- ▶ Small package for optimised PCB form factor
- ▶ USB for battery charging and data communication
- ▶ Flexible memory configuration option (flash/RAM) for over-the-air upgrade

—Franck Martins, MCU marketing and application, senior manager, GCS&A region at ST Microelectronics

terface in MCUs from Cypress, Texas Instruments, NXP and others. “While USB Type-C has just entered the market in late 2015, it is going to rapidly reach more than a billion consumers within the next 18 months—starting with smartphones, personal computers and tablets,” he says.

“USB Type-C is the biggest change in USB interface and, as described by many industry observers, it will soon be the one cable to rule these all. With its ease of use through reversible connections, higher power-carrying capability and flexibility to carry multiple types of data (video, USB, etc), it is going to pervade lots of electronics,” Sundaramoorthy adds.

Packing high-end performance

Baikal-T1 processor is the first Russian offering for the communications market to use a MIPS P-class Warrior CPU, boasting highly-competitive properties in terms of performance, technology node and compatibility.

Martins says, “Though the upper boundary level of MCU has been constantly pushed up in the past few years, thanks to higher speed

and processing power likewise, with the latest introduction of ARM cortex M7 (STM32F7) and STM32 enhanced architecture element such as ART accelerator, there is still a quantum barrier made up by the operating system (OS), which cannot be run out of MCU platform. Only a microprocessor unit (MPU) with a memory management unit (MMU) can operate iOS or Android OSes.”

Voica says, “We now have chips available, like M-class M5150 MCU, which integrate digital signal processing (DSP) and single instruction multiple data (SIMD) functionality, virtualisation support and advanced security features plus an MMU and cache controllers, which enable these to run full-fledged OSes like Linux.”

“Nevertheless, it is true that more functionalities previously handled by MPUs are managed by the MCU. For instance, the sensor-hub function (fusion of various environmental sensors data) that was implemented by the main AP (application processor or MPU) is now covered by a dedicated MCU with outstanding power-consumption benefit for battery-operated devices such as smartphones,” adds Martins.

Reducing manufacturing costs

“Broad technology changes in the last 18 months are mostly incremental and evolutionary such as continued lower power and better integration with more features like camera interfaces,” explains Sundaramoorthy.

Newer MCUs come with integrated functionality that was previously handled by external components. This allows design engineers to reduce their bill of materials (BoM). Availability of multi-standard support for communication standards, sensor fusion or hardware based security are all small but definite steps towards a more integrated, smaller and less expensive end system.

While cost savings are very minute on an individual level, these greatly add up with volume. Moreover, these simplify or solve many other problems faced while placing numerous components onto a circuit board that has to shrink with each successive generation to keep up with customer requirements.

“If everything is integrated together in a module, overall size and cost of the IoT solution will also be extremely less, thereby making products competitive. Engineers today are looking for MCU-integrated modules, where they can write their own applications,” explained Dhiraj Sogani, general manager and senior vice president, Systems Business Unit, at Redpine Signals Inc., in an interview with EFM. Redpine Signals has solutions coming up where an MCU will be integrated inside modules for vehicular communication.

The vision

The features we would like to see in more MCUs are in the area of vision capabilities. Adding computer vision to IoT MCUs will dramatically increase their applicability in broad areas including security/surveillance, drones, robotics and so on.

Key features needed in MCUs to support computer vision are integrat-